

SARANSH MITTAL

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Full-stack software engineer with experience in building AI-powered systems, real-time platforms, and scalable web applications. Experienced in LLM integration, multi-agent architecture, RAG pipelines, and event-driven backend systems. Strong in backend engineering, system design, and end-to-end product ownership through hands-on startup experience.

SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, C++ **Frontend:** React, Next.js (App Router), Tailwind CSS, Framer Motion, Web Audio API
Backend: Node.js, Express.js, FastAPI, REST API Design **Databases & Cache:** MongoDB (aggregation, indexing, vector search), PostgreSQL, Redis
AI / ML: OpenAI API, Claude API, LLM Integration, Multi-Agent Systems, Vector Embeddings, RAG, Semantic Search, Prompt Engineering
Infrastructure & Real-Time: Socket.io, SSE, Event-Driven Architecture, Docker, Cloudflare CDN, Railway, Web Push

EXPERIENCE

Software Engineer (Founding) — [Rapid Recap AI](#)

May 2024 –

Present

- Co-founded venture and led all engineering as primary developer across a 2-year lifecycle; collaborated with a cross-functional team of up to 5 (2 engineers, 2 product/business co-founders, 1 marketing) across multiple product phases
- Engineered AI-driven content pipeline using OpenAI APIs to auto-generate 10,000+ structured quizzes across 4 formats from real-world articles, reducing manual content work to zero
- Built real-time multiplayer matchmaking (1v1 and 4v4) with human-like bot simulation fallback, reducing average wait time from 3+ minutes to ~30 seconds under low-concurrency conditions
- Designed stateless backend with Redis session layer, connection-pooled MongoDB, and Cloudflare CDN; reduced TTFB to 200–300ms and load time from ~6–8s to ~3–4s via SSR and bundle optimisation
- Implemented semantic search using text embeddings and cosine similarity across 10,000+ articles; drove organic user acquisition via custom SEO pipeline with zero paid spend
- Designed asynchronous gameplay system with 24-hour challenge windows and independent per-player attempt timing, enabling cross-timezone participation and increasing session depth
- Built Python-based web scraping pipeline to replace delayed third-party news APIs; integrated scraped data into AI processing pipeline for real-time content summarisation and quiz generation

PROJECTS

HEALOSBENCH | *Python · Claude API · Haiku 4.5 · Prompt Engineering*

2025

- Built CLI eval harness to extract structured clinical JSON from synthetic medical transcripts using Claude Haiku 4.5; compared zero-shot, few-shot, and chain-of-thought prompt strategies with automated scoring across multiple output dimensions
- Designed field-level JSON diff grader with partial credit scoring and resumable run state; produced submission-ready academic paper and presentation documenting benchmark methodology and prompt engineering results

DiscipLog | *Next.js · TypeScript · MongoDB · OpenAI · Claude API · MCP · Docker*

2026

- Architected full-stack AI coaching platform: 7 subsystems including sprint timer, AI coaching workspace, RAG-based vector recall engine, analytics, developer activity ingestion, and gamified motivation engine
- Designed tool-calling AI coaching system with an LLM reasoning over structured tool outputs — vector log search, coach stats, domain analysis, belief recall — to generate evidence-grounded, context-aware coaching responses
- Built MCP server and VS Code extension for passive developer activity capture (IDE metadata, AI-agent turn tracking, sprint sync) and event-driven sessionizer pipeline converting raw events into structured logs with downstream memory compilation and coach card generation
- Engineered analytics layer with 12-week heatmaps, per-category progress tracking, and AI-generated weekly debriefs; implemented cron-driven coach card delivery queue with timing control to prevent notification fatigue

AI Yoga Guide | *Next.js · FastAPI · OpenAI · XTTS · SSE*

2025

- Built multi-agent AI system with triage agent dynamically routing user intent to specialised agents for breathing control, voice customisation, and settings management using tool-based execution
- Designed async XTTS voice generation backend (FastAPI + threading) with SSE-based real-time progress streaming; implemented low-latency Web Audio API playback synchronised with Framer Motion breathing animations

EDUCATION & ADDITIONAL

B.Tech in Computer Science — LNMIIT, Jaipur

2024

- 300+ LeetCode problems solved (160+ medium, 40+ hard, 95+ easy)
- Built and shipped 4 production applications end-to-end across AI systems, real-time gaming, and productivity — independently from ideation through deployment
- Completed structured self-study in system design: distributed rate limiting, consistent hashing, Kafka architecture, CQRS/Event Sourcing, Saga pattern, and CI/CD pipelines